

Section Report 1 — Sequence Similarity, Alignment, and Phylogeny

BIOL 4810

2026-02-17

Section Report 1

Sequence Similarity, Alignment, and Phylogenetic Inference

Purpose: Independently synthesize what you learned about BLAST, alignment, and phylogenetic inference.

Submission: Complete this template, upload required files, click **Prepare submission preview**, then use **Download as PDF (Print)**.

Part 1 — Dataset selection

Gene name:

ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit

Four organisms (one per line):

Vaucheria
Sargassum
Chlamydomonas
Fucus

How is this set different from class examples?

Although they are all algae like the previous sample, the gene I'm focusing on is different, and the type of algae I used has different characteristics. There are some coenocytic algae, green algae, brown algae, and a mix of

Part 2 — BLAST analysis

BLAST program used:

The NIH Nucleotide BLAST program

Two species selected:

Vaucheria
Sargassum

Interpret E-value, coverage, identity:

E-Value: My results came to an E-value of 0.0. This means that the similarity between my two algae is extremely close to each other and is unlikely to have occurred by chance.
Coverage: My coverage came out to 98%. This means that 98% of my sequence matches the sequences found in the data base.

One conclusion you can and cannot make:

One conclusion we can make is that the *Vaucheria* and *Sargassum* are very closely related due to the identity being 81.87% and the E-value being a 0. One conclusion we can not make is that how far ago these algae deviated from their common ancestor.

Upload BLAST screenshot

Choose File Screen Shot... 5.55.08 PM

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Job title: Nucleotide Sequence

RID [U1XFR92M114](#) (Expires on 02-28 08:40 am)

Query ID Icl|Query_2896503

Description None

Molecule type dna

Query Length 1467

Subject ID Icl|Query_2896505

Description None

Molecule type dna

Subject Length 1467

Program BLASTN 2.17.0+ [Citation](#)

Other reports: [Search Summary](#) [MSA viewer](#)

Graphic Summary

Distribution of the top 1 Blast Hits on 1 subject sequences ⓘ

Mouse over to see the title, click to show alignments

Color key for alignment scores

■ <40	■ 40-50	■ 50-80	■ 80-200	■ ≥200
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Query

View site information

Dot Matrix View

Descriptions

Sequences producing significant alignments:

Select: [All](#) [None](#) Selected: 0

[Alignments](#)
[Download](#)
[Graphics](#)

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Accession
☐	None provided	1415	1415	98%	0.0	81.87%	Query_2896505

Alignments Questions/comments

Part 3 — Multiple sequence alignment

Aligner used:

T-Coffee

Justification:

I chose T-Coffee for a few reasons. When we did the previous assignment, I had no issues with it and found it very easy/straight forward to work with. It didn't fight me on the documents I submitted, was pretty quick, and it

DNA alignment observations:

The first thing I notice about the DNA alignment is that the alignment result shows a lot of red coloring, meaning good alignments. There is some variable alignments in there too but, only a handful of bad.

Protein alignment observations (BLOSUM62):

I notice that, like the DNA alignment, the protein alignment shows a lot of red/good alignments, with some variable alignments and only a handful of bad alignments. Also, the protein and DNA alignments are very similar to each other.

Upload NESPrpt DNA PDF

Export/download the PDF from NESPrpt, then upload it here.

Choose File

0-0-1772158467-esp.pdf

f1fde...

1 / 3

75%

Vaucheria

Sargassum

Chlamydomonas

Fucus

1

10

20

30

NC

C.

VAUCHERIALITOREACHLOROPLASTCO

NC

C.

SARGASSUMTHUNBERGIIPLASTIDCOMI

NC

C.

CHLAMYDOMONASREINHARDTIIICHLOR

NC

C.

FUCUSVESICULOSUSCHLOROPLASTCOM

60

70

80

90

Vaucheria

Sargassum

Chlamydomonas

Fucus

CAAG

AA

GC

GT

ACT

CGA

ATT

AA

AA

AG

CCA

ACG

TT

AA

GG

AA

GA

AA

CT

CGA

AT

AA

AA

AG

TG

AG

CG

TT

AT

GA

CAC

AA

AC

AG

AA

ACT

AA

GC

AG

GT

GCT

GG

AT

TT

CA

AGG

AA

AG

AA

CT

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AT

TA

AA

AA

AG

TG

AG

CG

TT

AT

TC

120

130

140

150

Vaucheria

Sargassum

Chlamydomonas

Fucus

AT

GG

GT

TATT

GG

GAT

GC

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AG

AG

AT

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GGA

TATT

TG

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GC

AG

AT

TAC

AA

CG

TT

AA

AA

AC

Upload ESPrift Protein PDF

Export/download the PDF from ESPrift, then upload it here.

0-0-1772158550-esp.pdf

≡ 7e3d... 1 / 3 — 75% +       ⋮

Vaucheria GCAAAGCATATCGTG TAGATCCAGT
Sargassum GAAAGCATATCGTGTCGATCCTGTT
Chlamydomonas GTTGTTACGATATCGAACCAGTTCC
Fucus CGAAAGCATATCGTG TGGATCCTGTT

Part 4 — Phylogenetic analysis

Create a Maximum Likelihood tree using your protein alignment.

Conceptual explanation of Maximum Likelihood and how it differs from Neighbor-Joining:

Maximum likelihood uses the most probable events that occurred while Neighbor-Joining looks at the similarities between two groups and joins them that way.

What is bootstrapping?

Bootstrapping rebuilds the tree by resampling the columns. It counts how often clades appear in order to assume the relationship levels. Bootstrapping isn't a 100% accurate method, it's simply an assumption based on previous patterns

Interpret bootstrap values:

My bootstrap value was 100. This means that the program I used was very confident in the tree it gave me and that we can assume that the tree is extremely reliable.

Part 5 — Rooting

After midpoint rooting, describe any changes that have occurred in the tree:

When I midpoint rooted the tree, I didn't see any major changes. The only thing I noticed was that the little tail things changed their positions.

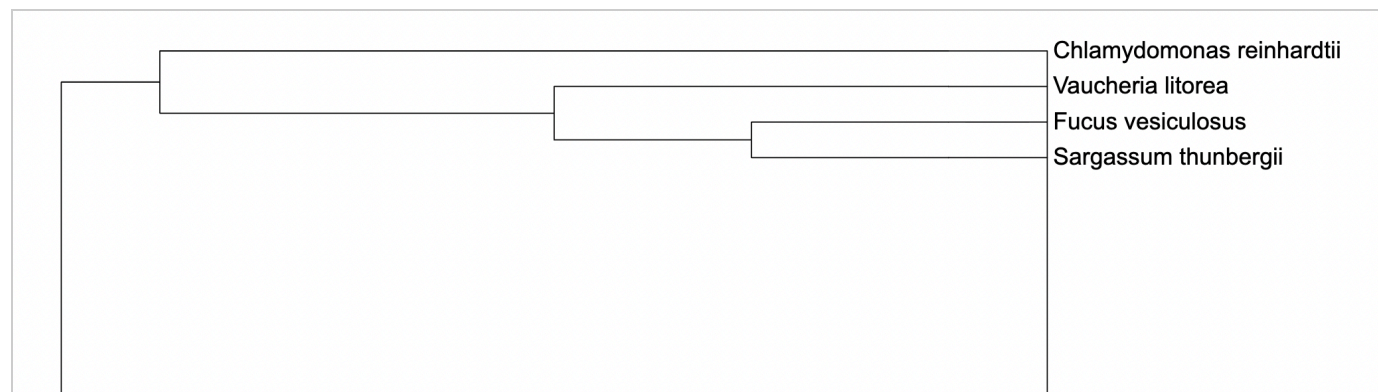
What assumption does midpoint rooting make and how is this different from using an outgroup?

Outgroup rooting uses a taxa that is known to not be a member of the main group in order to place a root between the outgroup and ingroup. Midgroup rooting is when a root is placed at the midpoint of the longest path

Upload iTOL UNROOTED screenshot

Choose File Screen Shot... 7.37.14 PM

The screenshot shows the iTOL web interface at itol.embl.de/tree/8216522107365581772163214. The main area displays a phylogenetic tree with four taxa: *Chlamydomonas reinhardtii*, *Vaucheria litorea*, *Fucus vesiculosus*, and *Sargassum thunbergii*. The tree is rooted at the midpoint. On the left, there are icons for zooming in/out, downloading, and editing. On the right, a 'Control panel' is open, showing various settings for the tree display. The 'Basic' tab is selected, and the 'Leaf sorting' is set to 'Default'. The 'Branch metadata display' section shows 'Node IDs', 'Bootstraps / metadata', and 'Range metadata' all set to 'Display'. The 'Node options' section shows 'Leaf node symbols' and 'Internal node symbols' set to 'Display'. The 'Auto collapse clades' section shows 'Auto collapse clades' set to 'cellular root (1)'. The 'Other functions' section shows 'Label functions' set to 'Multi-style', 'Auto assign taxonomy' set to 'NCBI', and 'Root the tree midpoint' set to 'Midpoint root'. At the bottom, there are buttons for 'Tree views', 'Undo', and 'Reset tree'.

Upload iTOL MIDPOINT-ROOTED screenshot Screen Shot... 7.36.18 PM**Part 6 — Synthesis**

Overall, what can be learned from doing this type of work? Where can uncertainty be introduced and how can that uncertainty be mitigated?

Aligning sequences allows us to compare DNA and protein in order to see how organisms are related and see where their similarities lie and differ. Also, focusing on similarities can give us the percent identity and alignment scores that arise from the inherited traits of a common ancestor. Phylogenies offer us hypothesis about evolutionary relationships based on the shared characteristics and inferred ancestry. It's important to remember that phylogenetic trees are inferences, not facts.

Export**Section Report 1 Submission Preview**

Part 1 — Dataset selection

Gene: ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit

Organisms:

Vaucheria
Sargassum
Chlamydomonas
Fucus

Novelty: Although they are all algae like the previous sample, the gene I'm focusing on is different, and the type of algae I used has different characteristics. There are some coenocytic algae, green algae, brown algae, and a mix of unicellular and multicellular algae in this sample.

Part 2 — BLAST analysis

BLAST program: The NIH Nucleotide BLAST program

Two species selected:

Vaucheria
Sargassum

Interpretation:

E-Value: My results came to an E-value of 0.0. This means that the similarity between my two algae is extremely close to each other and is unlikely to have occurred by chance.

Coverage: My coverage came out to 98%. This means that 98% of my sequence matches the sequences found in the data base.

Identity: My percent identity came out to 81.87%. This means that 81.87% of the amino acids are the exactly the same and in the exact same position.

Limits:

One conclusion we can make is that the Vaucheria and Sargassum are very closely related due to the identity being 81.87% and the E-value being a 0. One conclusion we can not make is that how far ago these algae deviated from their common ancestor.

BLAST screenshot:

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⚙

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Questions/comments

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Aligner: T-Coffee

Justification:

I chose T-Coffee for a few reasons. When we did the previous assignment, I had no issues with it and found it very easy/straight forward to work with. It didn't fight me on the documents I submitted, was pretty quick, and it was more conservative than Clustal Omega was.

DNA observations:

The first thing I notice about the DNA alignment is that the alignment result shows a lot of red coloring, meaning good alignments. There is some variable alignments in there too but, only a handful of bad.

Protein observations:

I notice that, like the DNA alignment, the protein alignment shows a lot of red/good alignments, with some variable alignments and only a handful of bad alignments. Also, the protein and DNA alignments are very similar to each other.

NESPrnt DNA PDF:

≡

f1fde...

1 / 3

—

75%

+

	1	10	20
Vaucheria	NC	C.VAUCHERIALITOREACHLORC	
Sargassum	NC	.SARGASSUMTHUNBERGIIPLAS	
Chlamydomonas	NC	C.CHLAMYDOMONASREINHARDT	
Fucus	NC	.FUCUSVESICULOSUSCHLORO	

	60	70	80
Vaucheria	CAAG	GC	TACTCGAATTAAAGCC
Sargassum	GGA	AAGA	ACTCGAATAAAGTGAGC
Chlamydomonas	CAC	AAAC	AGAACTAAAGCAGGTGCT
Fucus	AGG	AAAG	AACTCGATTAAAGAGTGAC

	120	130	140
Vaucheria	AT	GGGTTATT	GGGATGCATTTTATAA
Sargassum	GG	GATATTGG	GATGCAGATTACAAAGC
Chlamydomonas	CATA	CTACACAC	CTGATTACGTAGTAA
Fucus	TG	GATATTGG	GATGCAGATTACAAC

ESPrpt Protein PDF:



Part 4 — Phylogenetic analysis

Tree method: -

Method explanation:

Maximum likelihood uses the most probable events that occurred while Neighbor-Joining looks at the similarities between two groups and joins them that way.

Bootstrap explanation:

Bootstrapping rebuilds the tree by resampling the columns. It counts how often clades appear in order to assume the relationship levels. Bootstrapping isn't a 100% accurate method, it's simply an assumption based on previous patterns

Bootstrap interpretation:

My bootstrap value was 100. This means that the program I used was very confident in the tree it gave me and that we can assume that the tree is extremely reliable.

Part 5 — Rooting

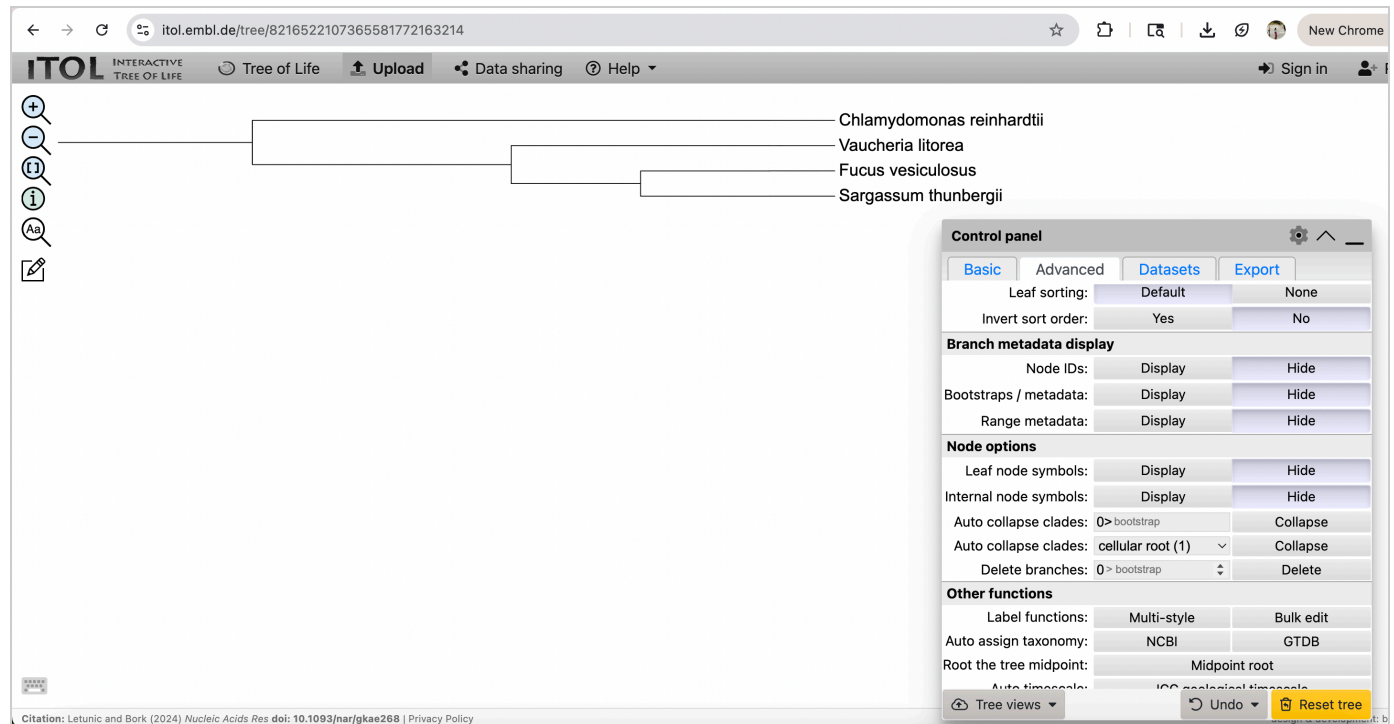
Unrooted vs midpoint-rooted comparison:

When I midpoint rooted the tree, I didn't see any major changes. The only thing I noticed was that the little tail thingies changed their positions.

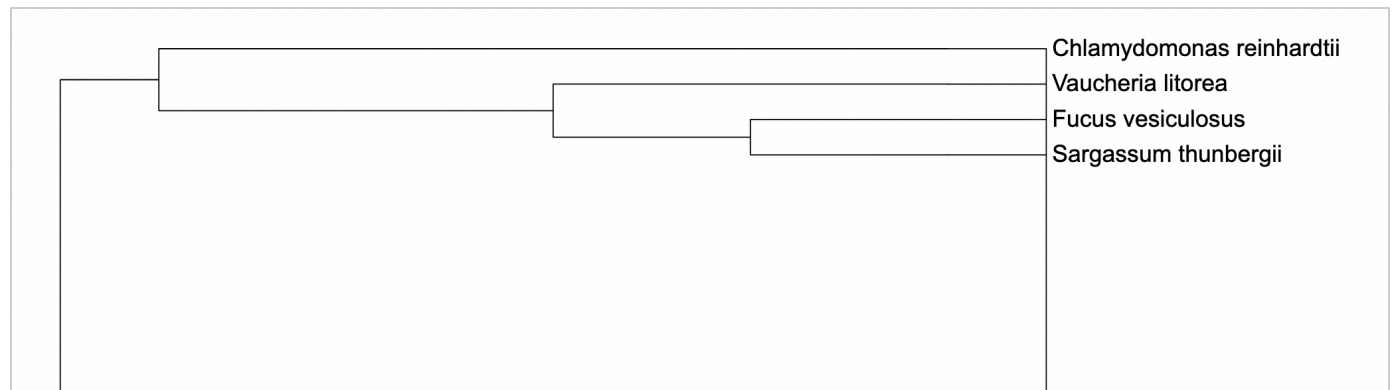
Midpoint rooting assumption:

Outgroup rooting uses a taxa that is known to not be a member of the main group in order to place a root between the outgroup and ingroup. Midgroup rooting is when a root is placed at the midpoint of the longest path of the tree in order to assume the approximate rates of evolution.

iTOL Unrooted screenshot:



iTOL Midpoint-rooted screenshot:



Part 6 — Synthesis

Aligning sequences allows us to compare DNA and protein in order to see how organisms are related and see where their similarities lie and differ. Also, focusing on similarities can give us the percent identity and alignment scores that arise from the inherited traits of a common ancestor. Phylogenies offer us hypothesis about evolutionary relationships based on the shared characteristics and inferred ancestry. It's important to remember that phylogenetic trees are inferences, not facts.